Al-Amin Hossain Nahid

Software Engineer | Full-Stack & AI Systems

Dhaka, Bangladesh | +8801305810184 | robin.nahid123@gmail.com

LinkedIn | GitHub | Portfolio

Professional Summary

Software engineer building full-stack applications and AI-powered systems, with professional experience across frontend architecture, backend services, agentic workflows and knowledge retrieval. Experienced in TypeScript, Next.js, Python, FastAPI, PostgreSQL, LLM tool calling, custom RAG architectures and cloud deployment. Focused on building reliable AI applications that combine conversational interfaces, structured workflows, document retrieval and executable tools.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, Kotlin, C/C++, C#
Frontend: React, Next.js, Tailwind CSS, shadcn/ui, Zustand, Plate.js, React Native
Backend: FastAPI, NestJS, ASP.NET Core Web API, REST API Design, Pydantic, SQLAlchemy, Prisma, TypeORM, Alembic, Authentication, Authorization
Databases & Storage: PostgreSQL, MySQL, MongoDB, Cloudflare R2
AI Engineering: LLM Integration, Agentic Workflows, Custom Agent Orchestration, Tool Calling, RAG, Knowledge Retrieval, Context Management, Structured Extraction, SSE Streaming
Infrastructure & Tools: Google Cloud Run, E2B, Docker, Git, Postman, Linux, Firebase, Vercel, Railway, Render

Professional Experience

KOLPO LTD *October 2026 – Present*
Software Engineer
Working on an enterprise AI application for a client, contributing across frontend development, backend services, agent orchestration, knowledge retrieval and cloud deployment.

- **Frontend Architecture:** Redesigned the application's user interface and developed full-stack features to improve usability and create a more consistent product experience.
- **Agentic Chat System:** Built a conversational AI system using custom Python orchestration and Pydantic-based models for structured data validation and agent interactions.
- **Knowledge Retrieval:** Developed a custom knowledge-base retrieval workflow that connects relevant organisational information to the AI agent to support context-aware responses.
- **Sandboxed Code Execution:** Integrated E2B to execute Python scripts in isolated sandbox environments and make execution results available to the agent workflow.
- **Cloud Deployment:** Worked with Google Cloud Run to deploy and operate the application in a production cloud environment.

AuraDev LTD *June 2026 – September 2026*
Software Engineering Intern
Contributed across the Next.js frontend and asynchronous FastAPI backend, building AI-assisted proposal workflows and knowledge management capabilities.

- **Agentic Proposal Generation:** Built a tool-driven workflow that guides users from grant source analysis and requirement extraction to proposal drafting, with human approval at key stages.
- **Conversational Agent & State Management:** Implemented an agentic chat experience, multi-step workflow orchestration, contextual interactions and conversation state management.
- **Custom Knowledge Retrieval:** Developed a summary-guided retrieval approach in which concise document descriptions stored in PostgreSQL help identify relevant files in Cloudflare R2 before retrieving their full contents for deeper analysis.
- **Collaborative Document Editing:** Integrated Plate.js commenting into the proposal editor to support document review and collaborative feedback.
- **Backend & Data Engineering:** Developed authenticated APIs and asynchronous backend functionality using FastAPI, SQLAlchemy, Alembic and PostgreSQL, with Clerk JWT authentication and organisation-based access control.
- **Document Workflows:** Contributed to PDF/DOCX processing, proposal version history and persistent document workflows.

Projects

Bilimbi Studio – Production Marketing Website Live | GitHub

- Built and deployed a content-driven website using Next.js, React, TypeScript and Tailwind CSS.
- Developed reusable UI components and design tokens and implemented a validated contact-form API with Zod and external email service integration.

AI Chat – Full-Stack AI Application Live | GitHub

- Developed a multi-conversation AI chat application with persistent history using Next.js, Prisma and PostgreSQL.
- Integrated Google Gemini through server-side API routes with streamed responses, conversation management and reusable React components.

MessMaster – Full-Stack Android Application GitHub

- Developed a native Android application for shared-living management, covering meal tracking, expense management, utility billing, notices and member communication.
- Built a NestJS and PostgreSQL backend with JWT authentication, role-based access control, validated APIs and TypeORM-based data persistence.
- Applied MVVM, Kotlin Coroutines, Retrofit and Material Design across the Android client.

Publications & Research

- **Al Amin Hossain Nahid**, Ferdus Hossain, Tariful Islam Fahim, and Salman Zzoha. "Intelligent Waste Classification for Sustainable Urban Development: A MobileNetV3-Based Deep Learning Approach Using the BDWaste Dataset." In *Proceedings of the International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII 2026)*, 2026. Published in IEEE Xplore.

Education

American International University-Bangladesh (AIUB) *September 2022 – October 2026*
B.Sc. in Computer Science and Engineering **CGPA: 3.65/4.00**